

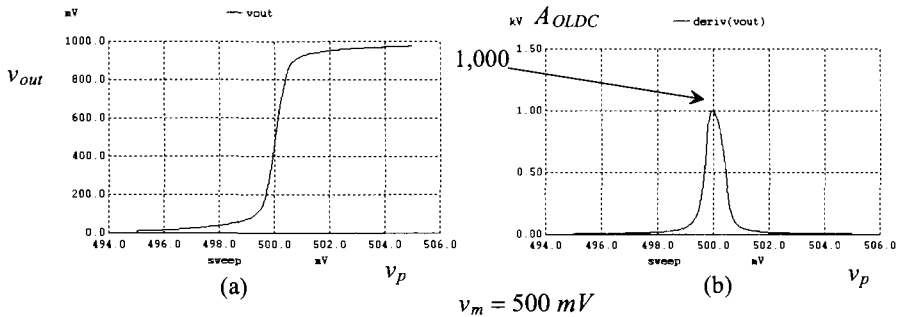


Figure 24.3 (a) DC transfer curves for the op-amp in Fig. 24.2 and (b) its gain (the derivation of (a)).

Power Dissipation

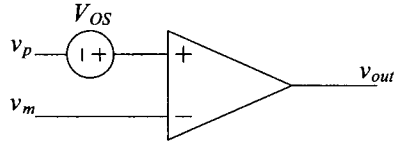
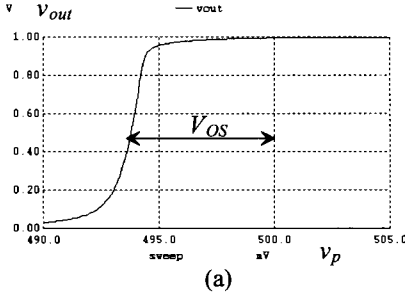
To determine the power dissipated by the op-amp, we sum the currents supplied by the constant current sources and multiply the result by V_{DD} . For the op-amp in Fig. 24.2, the current through M6 is 20 μ A and the current through M8 is 10 μ A. The total power dissipation is then 30 μ W ($V_{DD} = 1$ V).

Output Swing and Current Sourcing/Sinking Capability

The maximum output swing (for the op-amp in Fig. 24.2) is limited by M7 going into the triode region. If we must keep at least 100 mV across M7, then the maximum output voltage is 900 mV. When M8 goes into triode or roughly 100 mV (see Fig. 20.48), the minimum output voltage is set. As seen in Fig. 24.3a, the high-gain (or large slope) region falls between v_{out} of 100 and 900 mV. Note that the maximum amount of current that this op-amp can sink is limited by the constant current sink M8 or 10 μ A. The op-amp can source considerably more than 10 μ A by pulling the gate of M7 downwards. Because this topology can source a considerable amount of current, it is useful as a voltage regulator (because the op-amp is always only sourcing current in a voltage regulator application).

Offsets

We've talked in great length (earlier in the book) about random offsets and how to design and layout circuits to minimize their effects. Another type of offset (that is not random) is termed a *systematic offset*. When we sized the MOSFETs in Fig. 24.2, for example, we made sure that M7 was sized to source 10 μ A and M8 was sized to sink 10 μ A. What would happen if we sized M7 to source 100 μ A of current instead (changed its size from 100/2 to 1000/2)? Because M8 is a constant bias of 10 μ A, M7 would move into the triode region until it was sourcing the 10 μ A of current that M8 wants to sink. The output voltage would be very close to V_{DD} with M7 in the triode region. Effectively we would get a shift or offset in the transfer curves seen in Fig. 24.3a (see Fig. 24.4a). To model this output voltage shift, we can refer it back to the op-amp input as an input-referred offset voltage, Fig. 24.4b. Note that unlike a random offset, which may be positive or negative in value, a systematic offset will always be a known polarity. Finally, note that while we chose to model the offset in series with the noninverting input terminal, in Fig. 24.4, we could just as easily have placed it in series with the inverting op-amp terminal (with a change in polarity).



(b) How the offset is modeled

$$v_m = 500 \text{ mV}$$

Figure 24.4 Showing how increasing M7's width to 1000 in Fig. 24.2 causes an input-referred (systematic) offset voltage.

Compensating the Op-Amp

An important step in op-amp design is the design of the compensation network. The op-amp takes the difference between the inverting and noninverting input terminal voltages and, ideally, multiplies the result by a very large number (the gain). A block diagram representation of an op-amp is seen in Fig. 24.5. The open-loop gain as a function of frequency is labeled $A_{OL}(f)$.

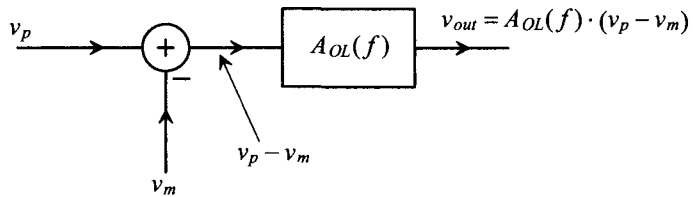


Figure 24.5 Block level diagram of an op-amp.

In all practical situations, the op-amp is used with feedback, Fig. 24.6. While our op-amp in Fig. 24.2 cannot drive a resistive load (unless it is megaohms, as a resistor on the op-amp output will kill the gain of the second stage), we'll still use this figure (Fig. 24.6) to illustrate the concept of feedback and compensation. We can write

$$v_{out} = A_{OL}(f) \cdot (v_{in} - v_f) \quad (24.2)$$

and

$$v_f = v_{out} \cdot \frac{R_2}{R_1 + R_2} \quad (24.3)$$

The amount of the output that is fed back is often called the feedback factor β or

$$\beta = \frac{R_2}{R_1 + R_2} \quad (24.4)$$

Substituting Eqs. (24.3) and (24.4) into Eq. (24.2) and solving for the closed loop gain gives

$$A_{CL}(f) = \frac{v_{out}}{v_{in}} = \frac{A_{OL}(f)}{1 + \beta \cdot A_{OL}(f)} \quad (24.5)$$

If we take $A_{OL}(f) \rightarrow \infty$, then the closed loop gain of this non-inverting topology is

$$A_{CL}(f) \rightarrow \frac{1}{\beta} = 1 + \frac{R_1}{R_2} \quad (24.6)$$

We have several important points that we need to discuss. To begin, notice that in Eq. (24.5) if

$$\beta \cdot A_{OL}(f) = -1 \quad (24.7)$$

or more precisely

$$|\beta \cdot A_{OL}(f)| = 1 \text{ and } \angle \beta \cdot A_{OL}(f) = \pm 180^\circ \quad (24.8)$$

the closed loop gain blows up (the feedback amplifier becomes unstable). The worst case situation (the largest value of β) occurs when all of the output is fed back to the op-amp input (assuming no transformer, amplifier, etc. in the feedback path). The voltage follower, Fig. 24.7, is an example of this situation. *To determine the stability of an op-amp, we'll look at the open loop gain when the feedback factor is one, that is,*

$$|A_{OL}(f)| = 1 \text{ and } \angle A_{OL}(f) = 180^\circ \quad (24.9)$$

Notice that the larger the closed loop gain, the smaller the value of β (the less output signal we feed back) and the more likely the op-amp circuit, with feedback, will be stable. *This is important.* While feedback helps to desensitize an amplifier's gain to variations in an op-amp's A_{OL} , the drawback is stability. In a high-performance op-amp that will never operate in a unity-follower configuration, we can get an enhancement in speed by reducing the amount of compensation (the amount of the reduction guided by Eq. (24.8) with the actual value of β used). While this discussion has focused on the non-inverting topology (voltage or series-shunt amplifiers, $\beta = v_f/v_{out}$) a similar discussion can be given for the inverting amplifier (shunt-shunt where $\beta = i_f/v_{out}$) topology, see Sec. 30.2.

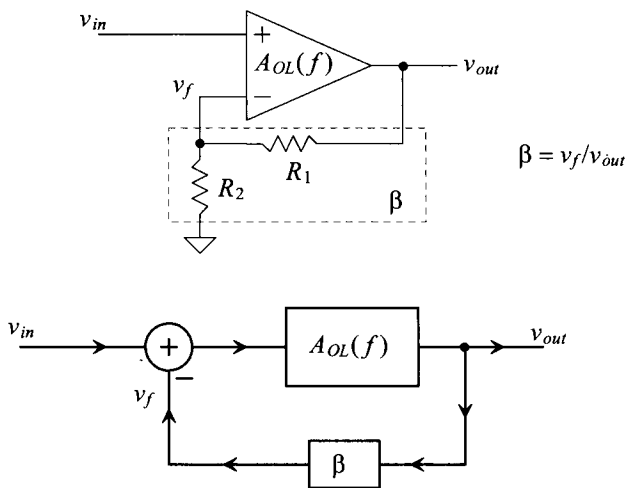


Figure 24.6 An example of feedback in an op-amp, see Sec. 30.2 for additional discussion.

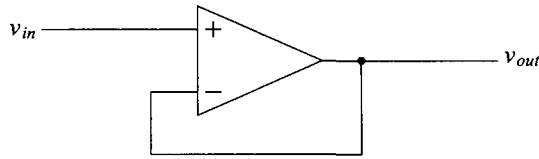


Figure 24.7 Voltage follower configuration, an example of a closed-loop amplifier with unity feedback factor.

To estimate the open-loop frequency response of the op-amp, let's use the generic model seen in Fig. 21.25. Figure 24.8 shows the location of nodes 1 and 2 on our two-stage op-amp. We've included a load capacitance in this figure. With the help of Table 9.2 we can write

$$R_1 = r_{on} || r_{op} = 111 \text{ k}\Omega$$

$$R_2 = r_{op} || R_{ocasn} \approx r_{op} = 333 \text{ k}\Omega$$

$$g_{m1} = g_{mn} = 150 \text{ }\mu\text{A/V (diff-amp)}$$

$$g_{m2} = g_{mp} = 150 \text{ }\mu\text{A/V (common-source)}$$

$$C_1 = C_{dg4} + C_{gd2} + C_{gs7} = 13.6 \text{ fF}$$

$$C_2 = C_L + C_{gd8} \approx C_L + 1.56 \text{ fF}$$

Let's calculate the open-loop response with a load and compensation capacitance of 100 fF, that is, $C_L = C_c = 100 \text{ fF}$. The pole associated with node 1, from Eq. (21.65), is

$$f_1 \approx \frac{1}{2\pi g_{m2} R_1 R_2 C_c} = 287 \text{ kHz}$$

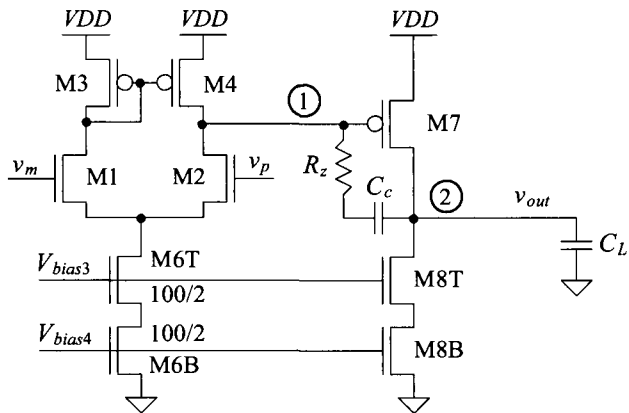


Figure 24.8 Calculating the frequency response of the op-amp.

The location of the pole associated with the output node (node 2) is, from Eq. (21.66),

$$f_2 = \frac{g_{m2}C_c}{2\pi(C_cC_1 + C_1C_2 + C_cC_2)} = 210 \text{ MHz}$$

The location of the zero, f_z in Eq. (21.63), is at 240 MHz (very near the second pole).

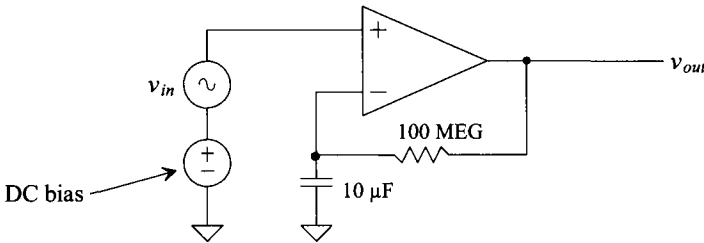


Figure 24.9 Circuit configuration used to simulate open-loop frequency response.

To simulate the open-loop response of the op-amp, we can use the configuration seen in Fig. 24.9. The feedback resistor and capacitor form a time constant so large that for all intents and purposes none of the AC output voltage is fed back to the inverting input. However, the DC bias level is fed back so that the op-amp biases up correctly (all MOSFETs are operating in the saturation region). With a DC bias voltage of 500 mV, Fig. 24.10 shows the open loop responses of the op-amp in Fig. 24.8 with 100 fF compensation capacitance and load capacitance. The simulated values are close to the

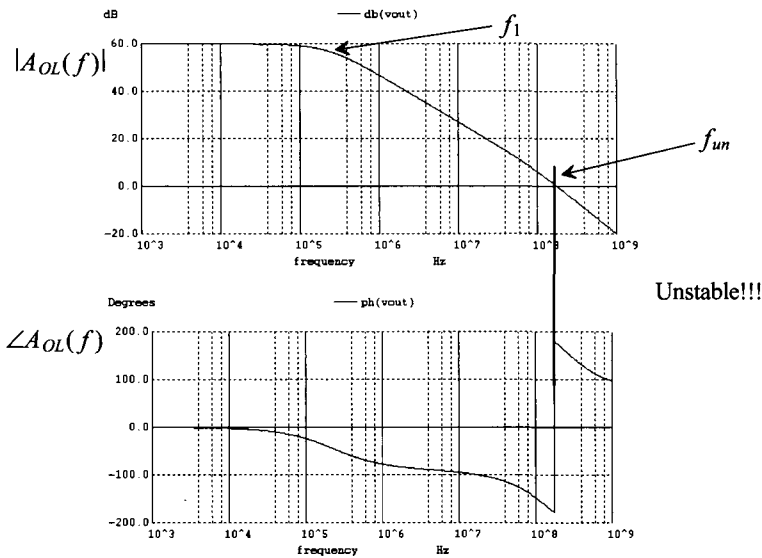


Figure 24.10 The open-loop frequency response of the op-amp in Fig. 24.8 with a 100 fF compensation capacitance and load capacitance.

hand-calculated values given above. However, from the criteria given in Eq. (24.9), the op-amp will clearly be unstable. *We want the open loop gain of the op-amp to be much less than one when the phase shift is 180° .* Looking at Fig. 24.10, we effectively want to shift the lower frequency pole (f_1) downwards in frequency. At the same time, we want to move the higher frequency pole, f_2 , higher in frequency (we want to *split the poles*). Remember this (pole-splitting) was the point of adding the compensation capacitor to an amplifier back in Ch. 21. Let's use Eq. (21.72) to select C_c . Looking at Fig. 24.10, we see that the phase shift is -100° at 10 MHz, so let's set the unity gain frequency to this value (in an attempt to make sure that the open loop gain is well under one when the phase shift is 180°).

$$f_{un} = \frac{g_{m1}}{2\pi C_c} = \frac{150 \mu A/V}{2\pi \cdot C_c} = 10 \text{ MHz} \rightarrow C_c = 2.4 \text{ pF} \quad (24.10)$$

The simulation results are seen Fig. 24.11. The unity-gain frequency, f_{un} , is close to 10 MHz. However, while the lower frequency moved downwards and the higher frequency pole moved upwards, the zero moved down to unity-gain frequency, see Eq. (21.63). This causes the open-loop gain to hover around unity instead of continuing to decrease (this is bad). The stability and step response of the op-amp will be degraded because of the zero. To illustrate this, consider the simulation results seen in Fig. 24.12. Notice that, in a transient simulation, we have to wait some time to let the bias circuit start up. Here, in this simulation, we've waited 500 ns before applying the input signal. Next, notice that our step input signal has a small amplitude change (5 mV). If we apply a larger step, the small-signal analysis used to derive pole-splitting is no longer valid (and we'll see the slew-rate limitations). Clearly the output of the op-amp doesn't behave well with the zero present. To eliminate the zero, we add the zero-nulling resistor, as seen in Fig. 21.33b. Figure 24.11 is regenerated in Fig. 24.13 with the addition of a zero-nulling resistor ($1/g_{mn} = 6.5k$). Figure 24.14 shows the well-behaved step response.

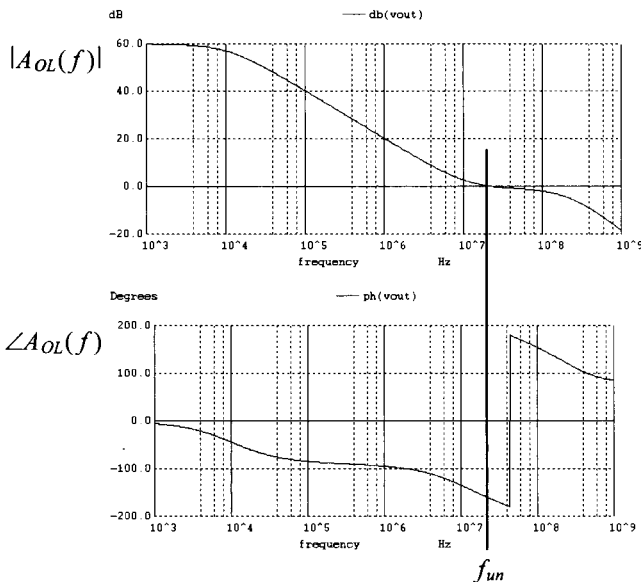


Figure 24.11 Increasing the compensation capacitor's value to 2.4 pF.

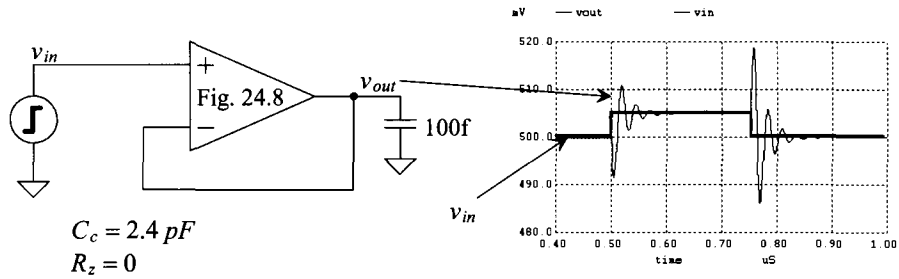


Figure 24.12 The poor step response of the op-amp with the zero present.

Gain and Phase Margins

In the previous discussions we assumed a load of 100 fF. If the load capacitance varies, the stability can (will) be affected. Further, with temperature, process, and power supply variations, the stability of the op-amp can change. In order to specify “how stable” the op-amp is at a given set of operating conditions, the parameters gain margin (GM) and phase margin (PM) are used. To determine an op-amp’s PM, we look at the phase shift when the open-loop gain is unity. The amount of phase shift away from 180° is the PM of the op-amp. As seen in Fig. 24.13, the phase shift when the gain is unity is -90° . Taking the difference between this value and 180° gives a PM of 90° . To calculate the GM, we look at the difference between the open-loop gain and unity when the phase of the op-amp is $\pm 180^\circ$. For the op-amp response in Fig. 24.13, the GM is approximately 25 dB. Note that as seen in Fig. 24.14, it is nice to have a PM of 90° because the step response of the op-amp has a first-order response (like an RC circuit).

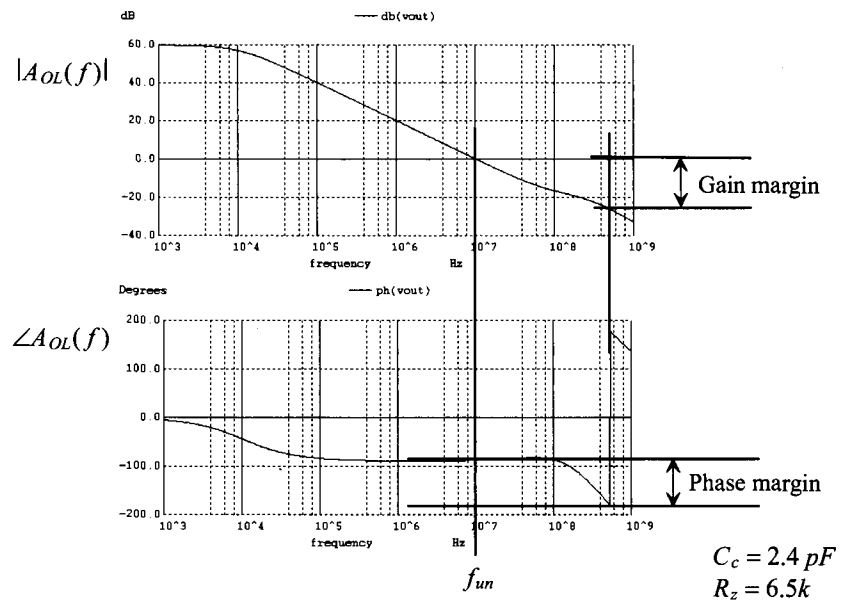


Figure 24.13 Adding a zero nulling resistor to the op-amp in Fig. 24.8.

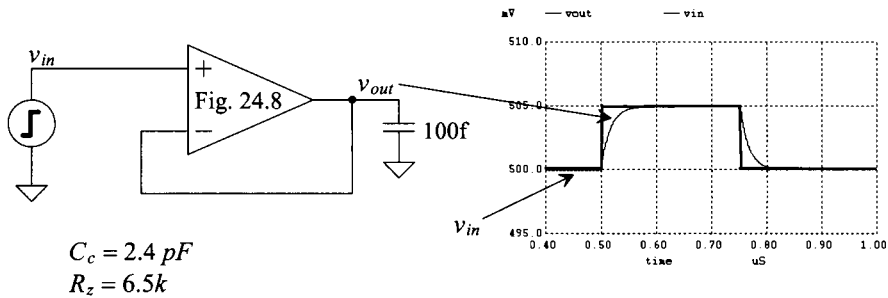


Figure 24.14 Good step response of the op-amp with the zero absent.

Removing the Zero

As the previous discussion illustrated, the RHP zero can raise issues concerning stability and settling time (this is important). As seen in Eq. (21.73) and the associated discussion, R_z can be added to eliminate ($R_z = 1/g_{m1}$) or move the zero into the LHP ($R_z > 1/g_{m1}$). When the zero is moved to the LHP, the phase response of the zero adds to the overall phase response, increasing the PM (this is called *lead compensation*). The practical problem with using R_z is that setting its value to a precise number (say $1/g_{m1}$) is challenging with shifts in process, temperature, or voltage. One solution to this problem is to replace the resistor with a MOSFET operating in the triode region, Fig. 24.15. M_z behaves like a resistor with a value of $1/g_{m1}$. Ideally, the source-gate voltage of M_7 is the same as the V_{SG} of $MP1$. It then follows that the source-gate potential of $MP2$ equals the source-gate potential of M_z . The channel resistance of M_z is then, from Eq. (9.16), set to $1/g_{m1}$. The practical issues with this method are the wasted power dissipated by the additional circuitry and the fact that if the output swing becomes large (especially at higher frequencies where C_c has a small impedance), M_z can move out of the triode region, which can affect the large-signal behavior of the op-amp.

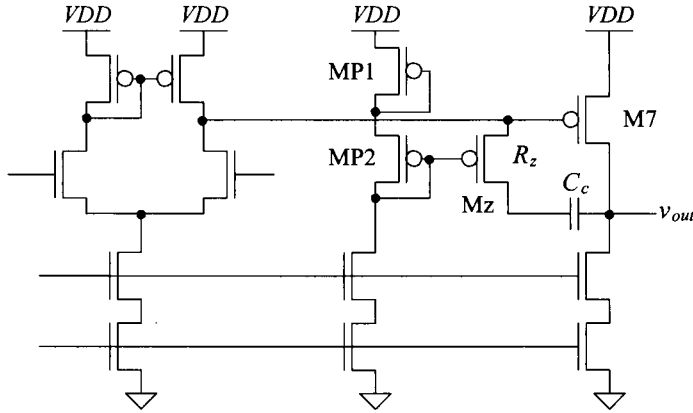


Figure 24.15 Making the zero-nulling resistor process independent.

The other method to remove the RHP zero seen in Fig. 21.33a is to add an amplifier with a gain of +1 in series with the compensation capacitor. Figure 24.16 shows the idea. A source follower allows the output signal to feed back through the compensation capacitor (so that the pole-splitting effect is still present). However, the root cause of the RHP zero, namely, C_c shunting the input of the second stage (the output of the diff-amp) to the output of the second stage (the output of the op-amp) at higher frequencies is removed. The concerns with this topology are, again, power dissipation and, perhaps more importantly, large signals. If the overall output voltage swings too low, it causes the source-follower to shut off.

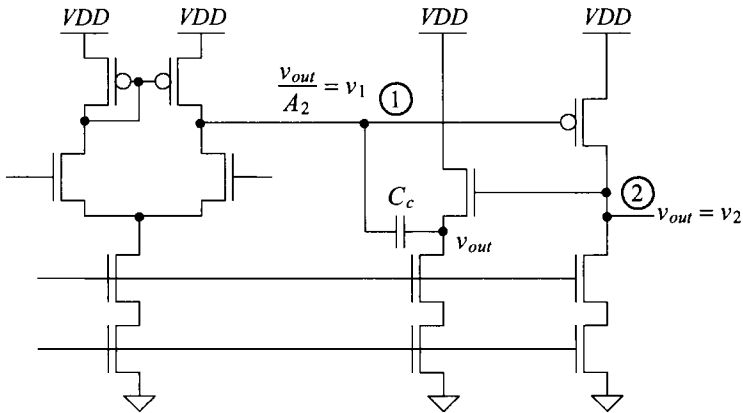


Figure 24.16 Using an amplifier to eliminate forward signal feedthrough via the compensation capacitor.

Compensation for High-Speed Operation

Notice, in Fig. 24.16 (or in any other of the topologies we've discussed up to this point), that the current fed back through C_c is

$$i_{Cc} = \frac{v_{out} - \frac{v_{out}}{A_2}}{1/j\omega C_c} \quad (24.11)$$

If the second-stage gain, A_2 ($g_{m2}R_2 = g_{m7}r_{o7}$) is reasonably large, then this equation can be approximated using

$$i_{Cc} \approx \frac{v_{out}}{1/j\omega C_c} \quad (24.12)$$

If we can feed back this current *indirectly* to the output of the diff-amp, we can still compensate the op-amp (and have pole-splitting). Further, if we do it correctly, we avoid connecting the compensation capacitor directly to the output of the diff-amp and thus avoid the RHP zero. Towards this goal, consider the modified op-amp schematic seen in Fig. 24.17. The added MOSFETs form a common-gate amplifier. (Note that here we are assuming $v_p \ll v_{out}$, so the source-follower action of MCG is negligible. MCG is connected to v_p to set its gate at the DC voltage of the input.) The current i_{Cc} is fed back through the common-gate MOSFET, MCG, to node 1 (the output of the diff-amp).

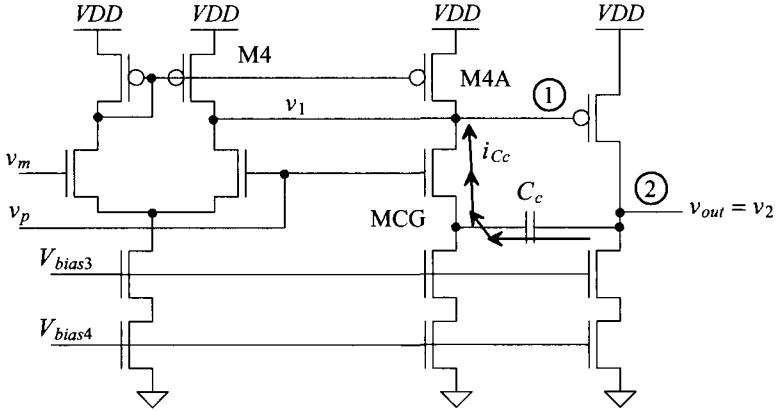


Figure 24.17 Feeding back a current indirectly to avoid the RHP zero.

To determine the frequency response of this amplifier, consider the model seen in Fig. 24.18 (modified from Fig. 21.25). Summing the currents at node 1 gives

$$-g_{m1}v_s + \frac{v_1}{R_1 \parallel \frac{1}{j\omega C_1}} - \frac{\overbrace{v_{out}}^{i_{Cc}}}{1/j\omega C_c + 1/g_{mcg}} = 0 \quad (24.13)$$

where the resistance looking into the source of MCG is $1/g_{mcg}$. Solving for v_1 gives

$$v_1 = \frac{g_{m1}R_1}{1 + j\omega R_1 C_1} \cdot \left(\frac{j\omega \frac{C_c}{g_{m1}}}{1 + j\omega \frac{C_c}{g_{mcg}}} \cdot v_{out} + v_s \right) \quad (24.14)$$

For the output node (node 2), we can write (noting that now C_2 includes both C_c and C_L)

$$v_{out} = -g_{m2}v_1 \cdot \left(\frac{R_2}{1 + j\omega R_2 C_2} \right) \quad (24.15)$$

Plugging Eq. (24.14) into this equation gives

$$v_{out} = -g_{m1}R_1g_{m2}R_2 \cdot \frac{\frac{j\omega \frac{C_c}{g_{m1}}}{1 + j\omega \frac{C_c}{g_{mcg}}} \cdot v_{out} + v_s}{(1 + j\omega R_1 C_1)(1 + j\omega R_2 C_2)} \quad (24.16)$$

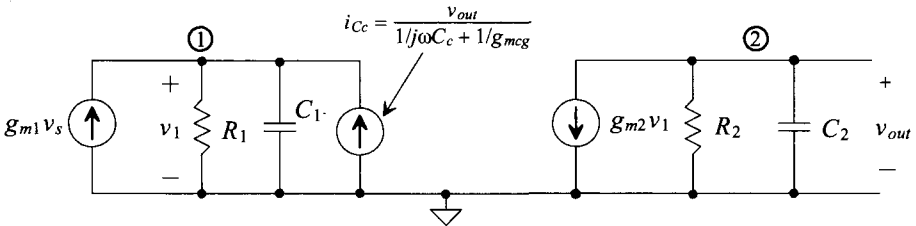


Figure 24.18 Model used to estimate bandwidth when indirect feedback current is used.

Letting

$$K = \frac{-g_{m1}R_1g_{m2}R_2}{(1+j\omega R_1C_1)(1+j\omega R_2C_2)} \quad (24.17)$$

we can rewrite Eq. (24.16) as

$$v_{out} = \frac{j\omega \frac{C_c}{g_{m1}}}{1+j\omega \frac{C_c}{g_{mcg}}} \cdot K \cdot v_{out} + K \cdot v_s \quad (24.18)$$

and, assuming $|K \cdot \frac{C_c}{g_{m1}}| \gg \frac{C_c}{g_{mcg}}$, thus

$$\frac{v_{out}}{v_s} \approx \frac{K \cdot \left(1 + j\omega \frac{C_c}{g_{mcg}}\right)}{1 - j\omega \frac{C_c K}{g_{m1}}} \quad (24.19)$$

Resubstituting in K gives

$$\frac{v_{out}}{v_s} = \frac{-g_{m1}R_1g_{m2}R_2 \left(1 + j\omega \frac{C_c}{g_{mcg}}\right)}{(1+j\omega R_1C_1)(1+j\omega R_2C_2) + j\omega \frac{C_c}{g_{m1}} \cdot g_{m1}R_1g_{m2}R_2} \quad (24.20)$$

or, with $s = j\omega$, we can write

$$\frac{v_{out}}{v_s} = \frac{-g_{m1}R_1g_{m2}R_2 \left(1 + s \frac{C_c}{g_{mcg}}\right)}{s^2 \cdot (R_1C_1R_2C_2) + s \cdot (R_1C_1 + R_2C_2 + R_1g_{m2}R_2C_c) + 1} \quad (24.21)$$

Notice there is a LHP zero at

$$f_z = \frac{g_{mcg}}{2\pi C_c} \quad (24.22)$$

Since this zero is in the LHP, it will add to the phase response and enhance the speed of the op-amp. Intuitively, we can think that at high speeds the phase shift through C_c will cause the output signal to feed back and add to the signal at node 1. This *positive feedback* enhances the speed of the op-amp. To determine the location of the second pole, let's assume $R_1g_{m2}R_2C_c \gg R_1C_1$ or R_2C_2 or $R_1C_1R_2C_2$ so

$$s_{1,2} \approx \frac{-R_1g_{m2}R_2C_c \pm R_1g_{m2}R_2C_c}{2(R_1C_1R_2C_2)} \quad (24.23)$$

Our approximation won't tell us the location of the lower frequency pole (it's given by Eq. (21.65)). The location of the second pole is

$$f_2 \approx \frac{g_{m2}C_c}{2\pi \cdot C_1C_2} \approx \frac{g_{m2}C_c}{2\pi \cdot C_1(C_L + C_c)} \quad (24.24)$$

This result should be compared to Eq. (21.66). The location of the second pole is at a considerably higher frequency using this technique. *The result is that we can set the unity gain frequency to a higher value and still have a stable op-amp.* Further, the load capacitance (which is included in C_2) can be considerably larger for a given PM or GM. Again, Eq. (21.72) can be used to set the unity-gain frequency, f_{un} , assuming $f_2 \approx f_z$

$$f_{un} = \frac{g_{m1}}{2\pi C_c} (\approx f_z \text{ if } g_{m1} \approx g_{mcg}) \quad (24.25)$$